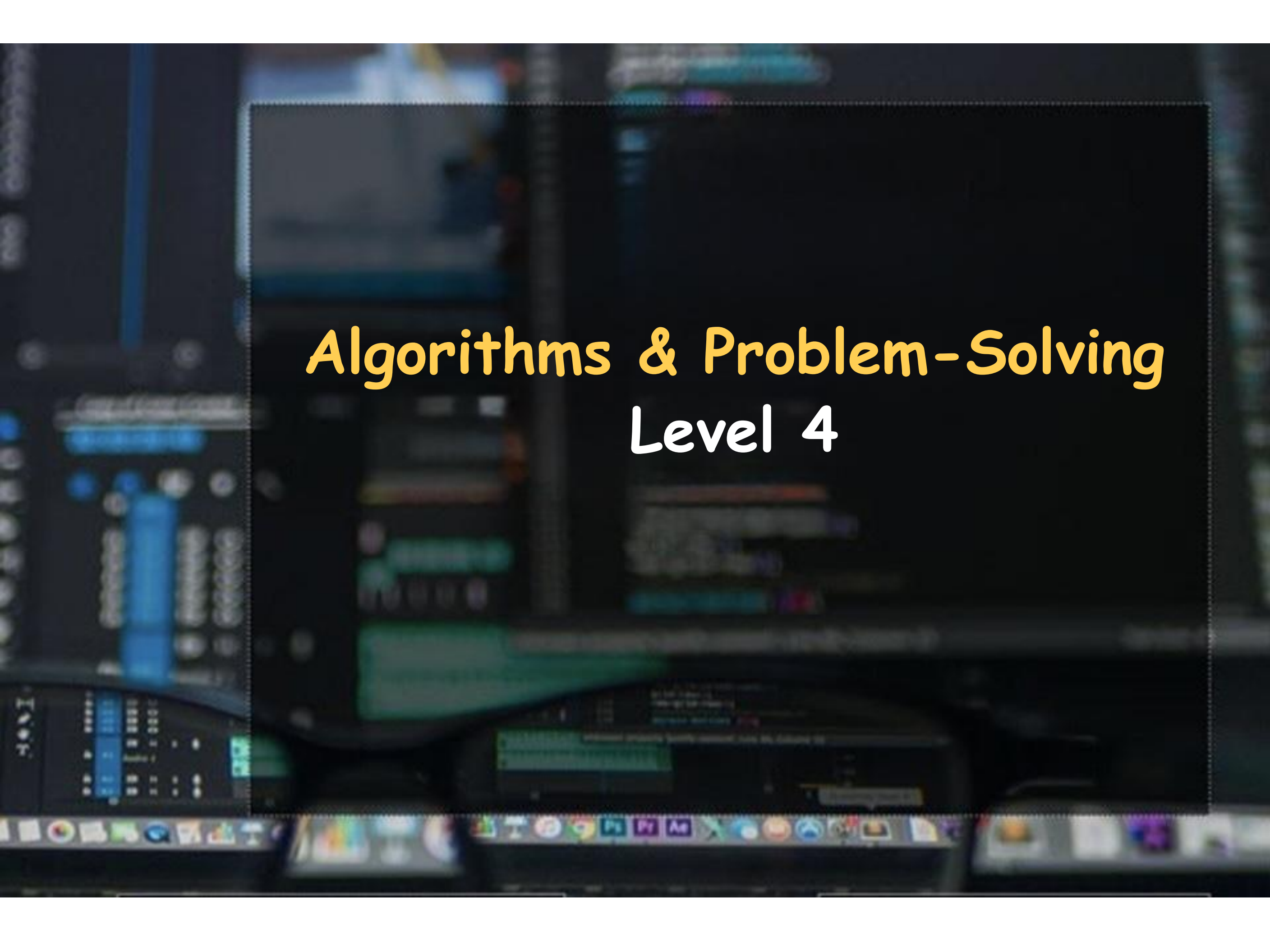




Algorithms & Problem-Solving Level 4

The background of the slide features a complex network diagram with numerous nodes and connecting lines, overlaid on a dark, semi-transparent rectangular area. In the lower portion of the image, the silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, looking at a screen or document.

Algorithms & Problem Solving

Level 4

Date1 Less Than Date2

Definition

-  The goal is a program that checks whether the first date (Date1) precedes the second date (Date2) in time.



Explanation

■ The program uses a data structure to store each date as three elements: day, month, and year.

■ After the user enters the two dates, the years are compared first, then the months, and finally the days to determine which is older.

■ The comparison method in the `IsDate1BeforeDate2` function:

- ① If the year of the first date is younger → the first date is older.
- ② If the two years are the same → the months are compared.
- ③ If the months are also the same → the days are compared.

Procedures

- 1 Read the first date:
 -  Using the `ReadFullDate()` function, which retrieves the day, month, and year.
- 2 Read the second date:
 -  Using the same method as above.
- 3 Perform the comparison:
 -  Using `IsDate1BeforeDate2(date1, date2)` to determine which date is earlier.
- 4 Display the result:
 -  Print a message indicating whether the first date is earlier than the second.



Example:

Please enter a Day? 12

Please enter a Month? 2

Please enter a Year? 2022

Please enter a Day? 12

Please enter a Month? 3

Please enter a Year? 2022

Yes, Date1 is Less than Date2.



Note:

- The comparison is accurate because it starts with the year and ends with the day.
- The `IsDate1BeforeDate2` function returns a Boolean value (true/false).
- The program does not check the date (e.g., February 30th).

Summary

- ✓ The program allows for accurate chronological verification of the order between two dates.
- ✓ The method used is simple and logical, and is fundamental to applications such as date sorting or calculating time differences.

Thank you 😊